Practical handoff and recovery

Saved simulation evidence | 24 October 2026

100 completed seeded runs: two plants, five families, SG/plain, five seeds.
3,733 paired E1 decisions: 3,200 polymer + 533/640 column (frozen 22:36 UTC).
19 figures; complete data, SHA-256 provenance and independent verification.

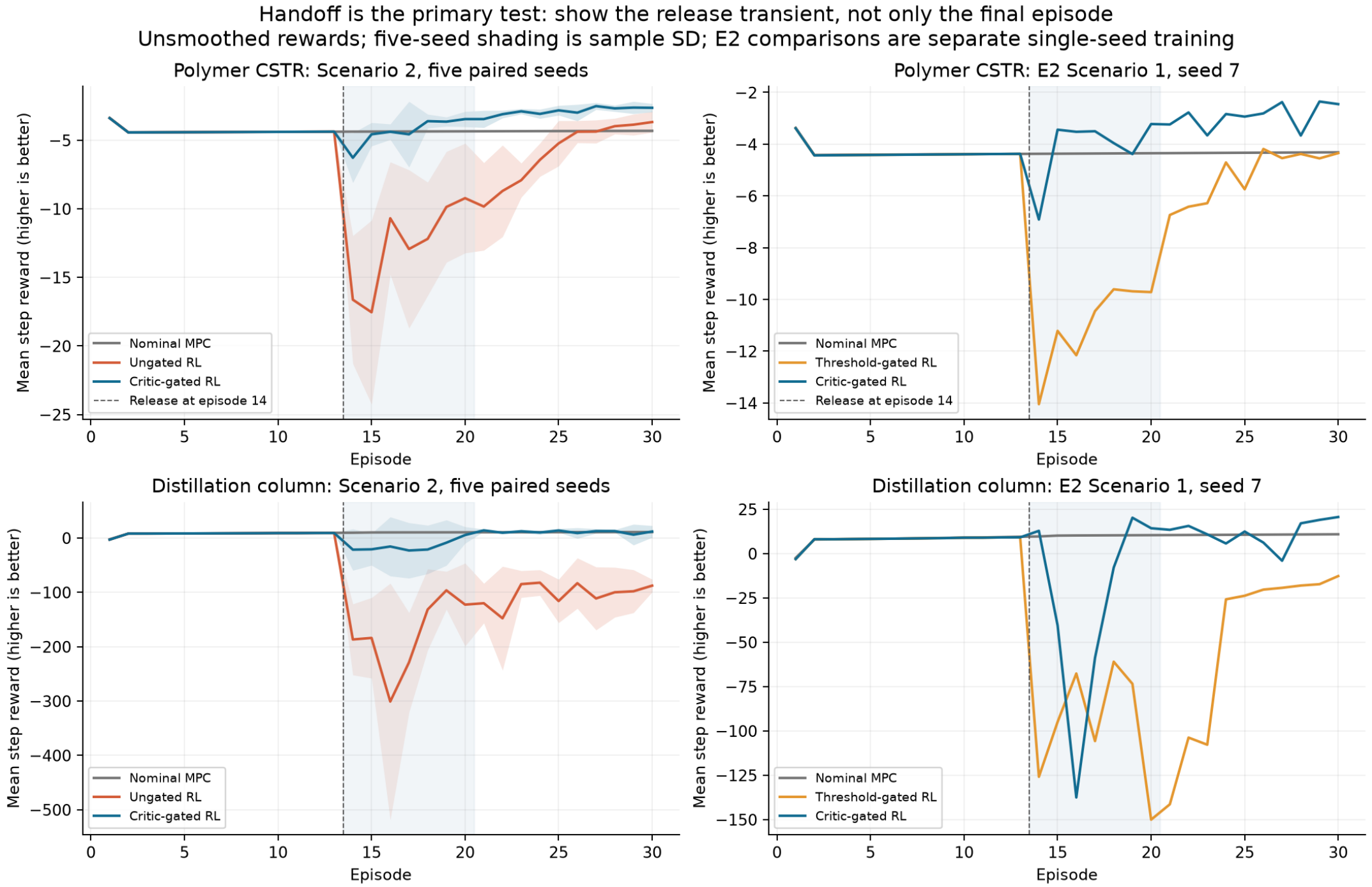
Combined SG reduces handoff disruption compared with ungated RL in all five paired seeds in both plants. Benefits are strongest for residual and combined.

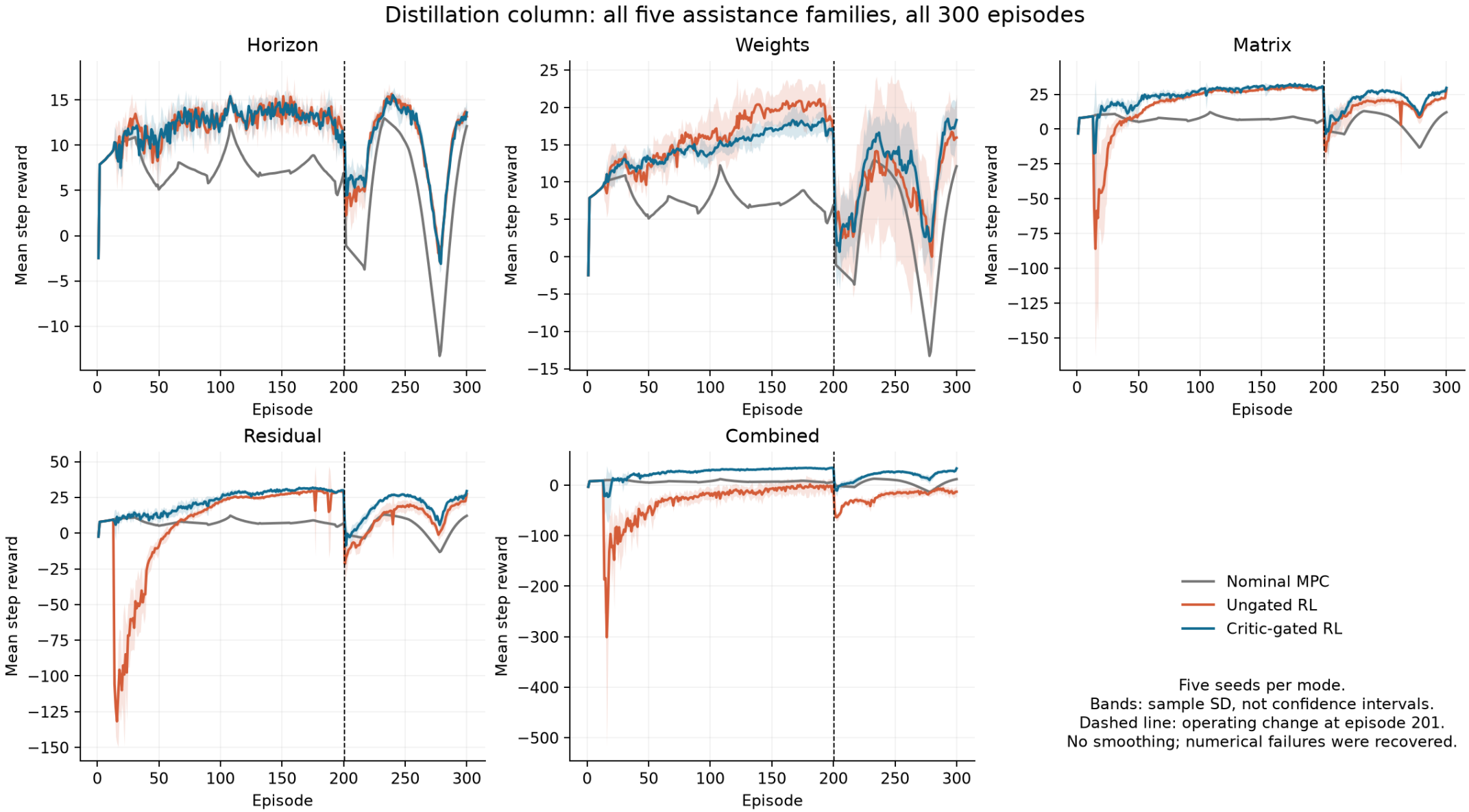
Median first recovery of BOTH output RMSEs to matched MPC (three episodes):
- Polymer: SG 5 episodes; ungated RL 19.
- Column: SG 24; ungated RL 78 (one ungated seed not recovered by episode 200).

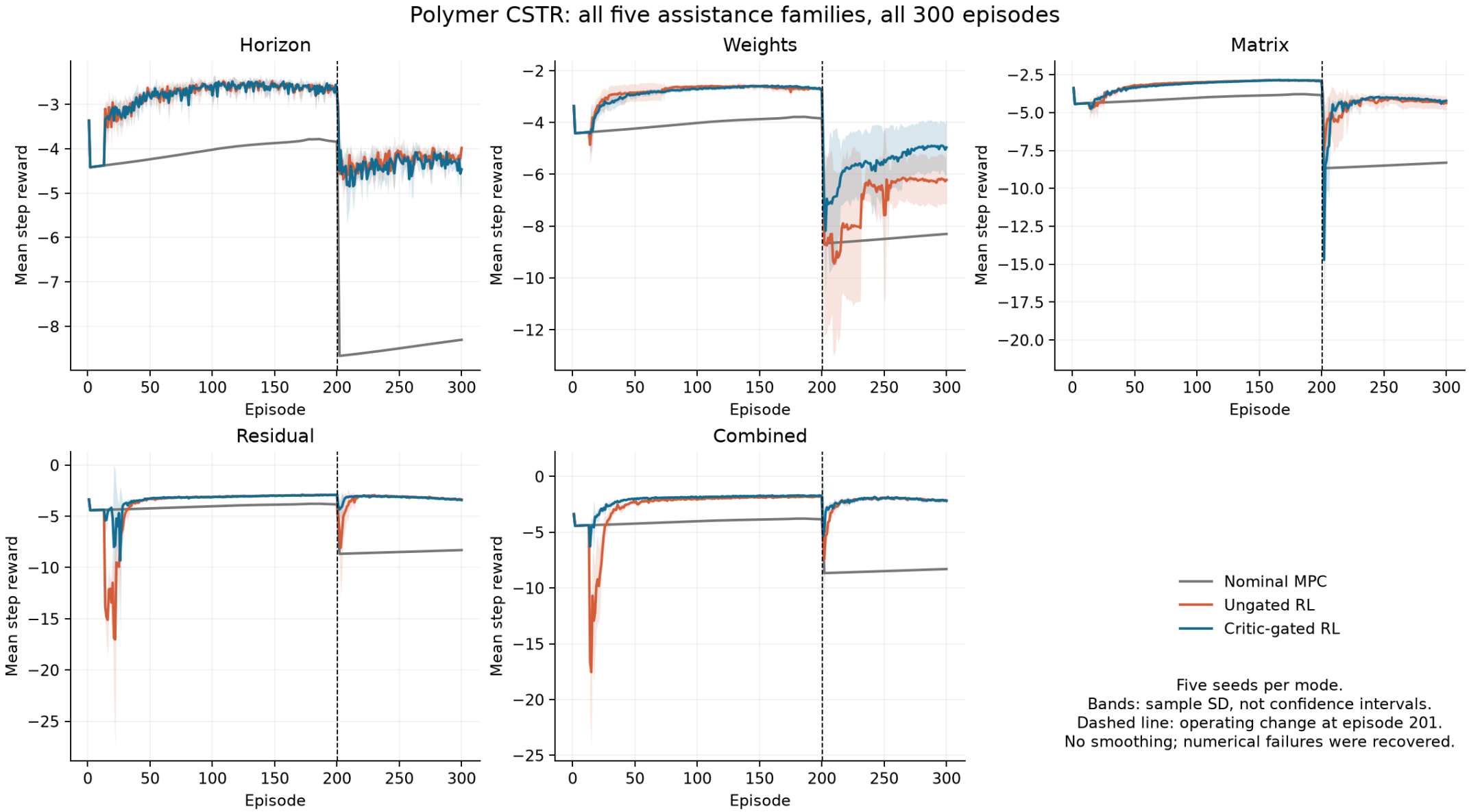
Different horizon/weight settings are often nearly equivalent near steady state. Supervisor retention in those cases varies by plant and channel.

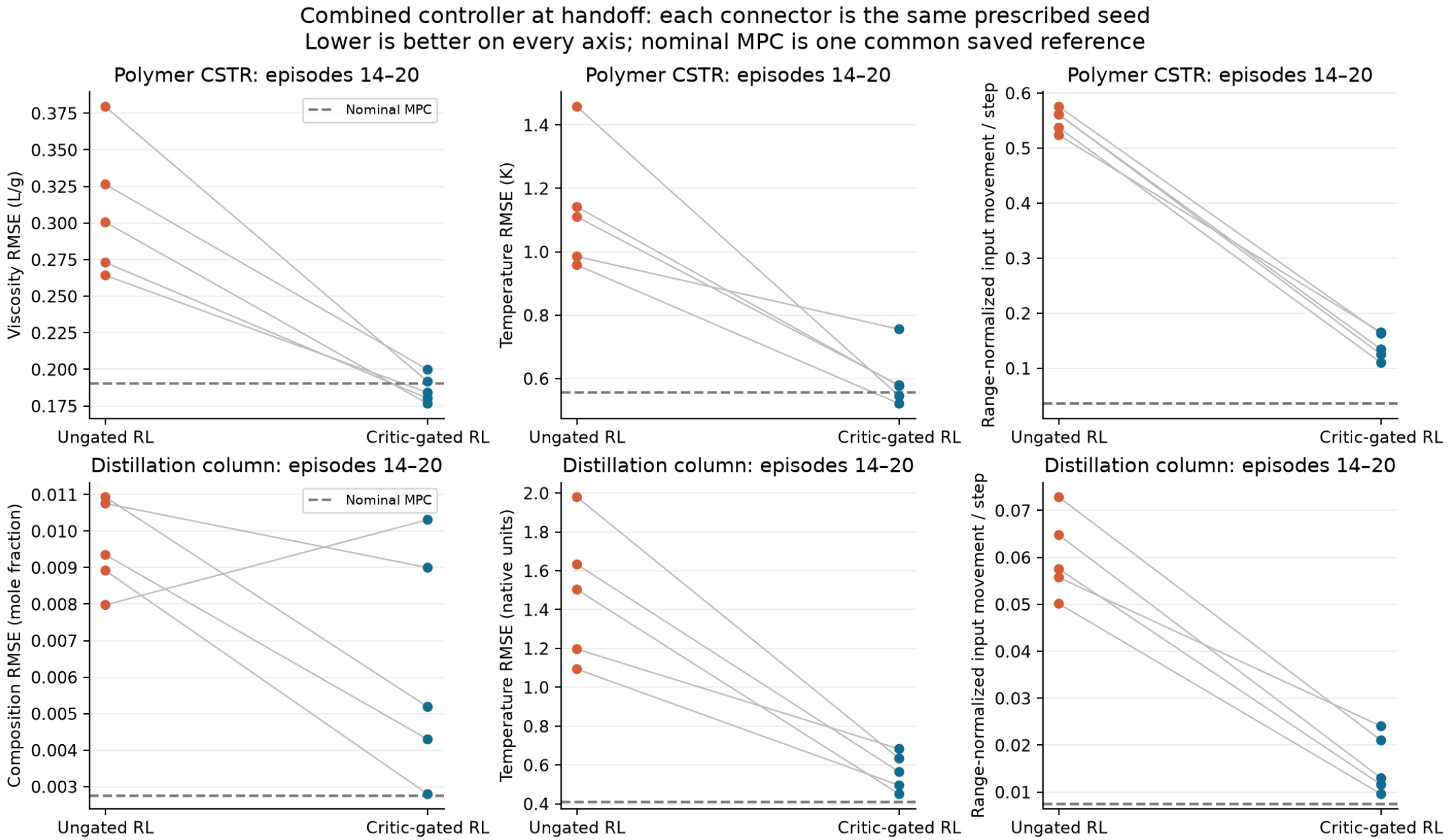
Counterexamples are retained: threshold can win individual recovery metrics; late polymer performance and E1 critic decisions are not uniformly better.

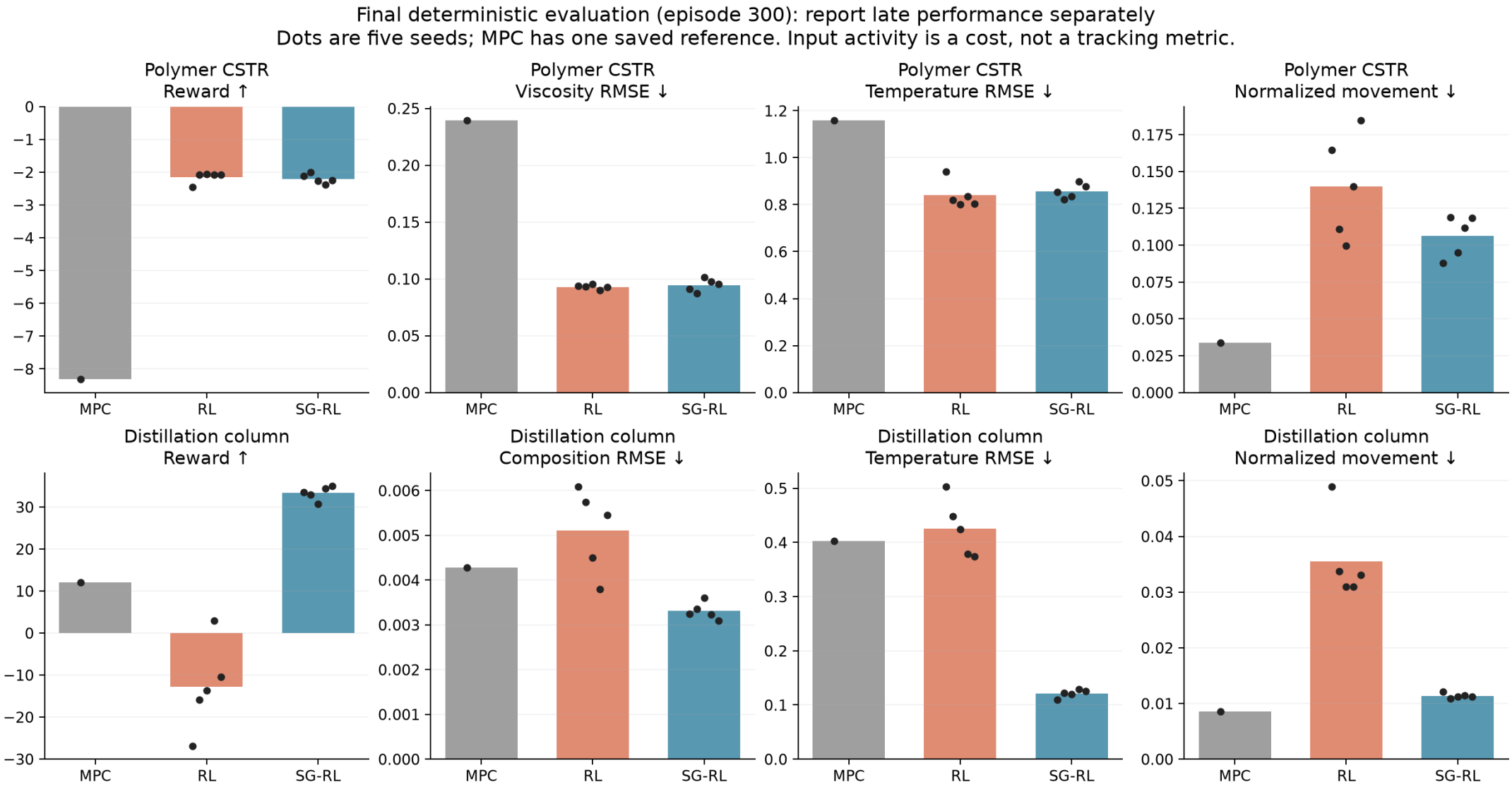
No new simulation or manuscript/reviewer-response edit. Raw data and recovery checkpoints remain in their verified original locations and lossless archives.
Full definitions, per-seed results and interpretation: ASSESSMENT.md and data/.


Handoff is the primary test: show the release transient, not only the final episode
Unsmoothed rewards; five-seed shading is sample SD; E2 comparisons are separate single-seed training
Polymer CSTR: Scenario 2, five paired seeds
Polymer CSTR: E2 Scenario 1, seed 7
Distillation column: Scenario 2, five paired seeds
Distillation column: E2 Scenario 1, seed 7
Mean step reward (higher is better)
Episode
Nominal MPC
Ungated RL
Threshold-gated RL
Critic-gated RL
Release at episode 14

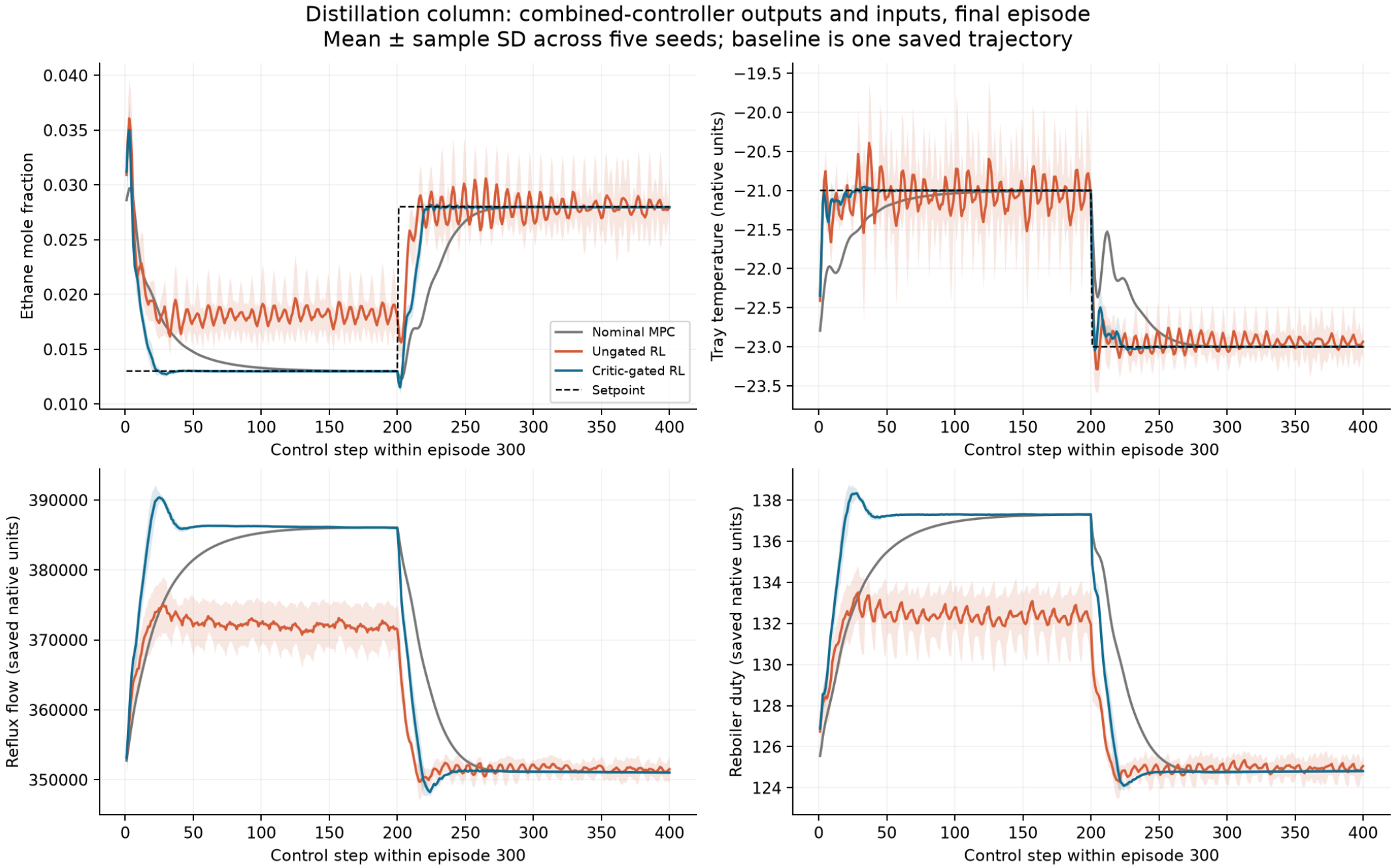

Distillation column: all five assistance families, all 300 episodes
Horizon
Weights
Matrix
Residual
Combined
Mean step reward
Episode
Nominal MPC
Ungated RL
Critic-gated RL
Five seeds per mode.
Bands: sample SD, not confidence intervals.
Dashed line: operating change at episode 201.
No smoothing; numerical failures were recovered.

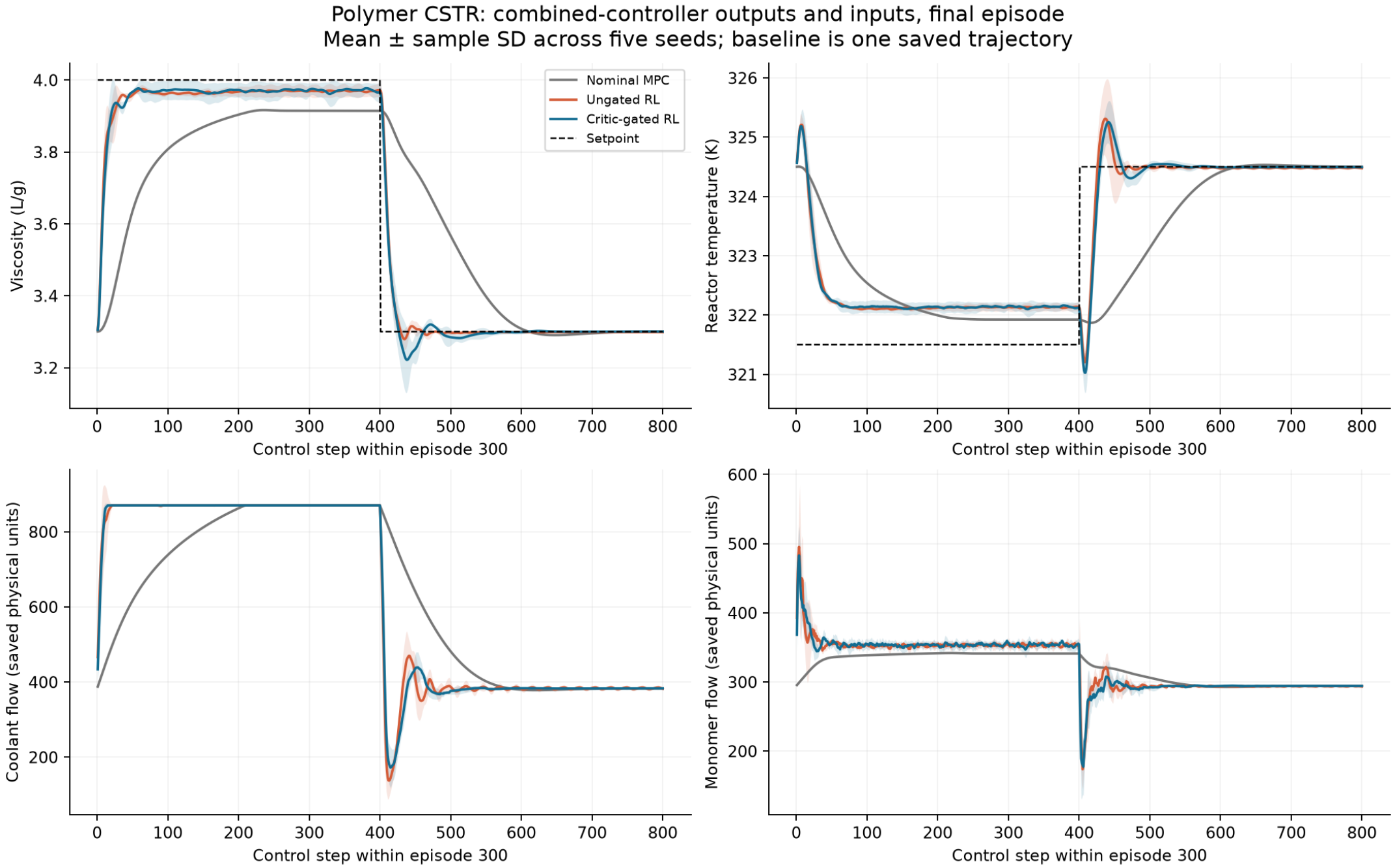

Polymer CSTR: all five assistance families, all 300 episodes
Horizon
Weights
Matrix
Residual
Combined
Mean step reward
Episode
Nominal MPC
Ungated RL
Critic-gated RL
Five seeds per mode.
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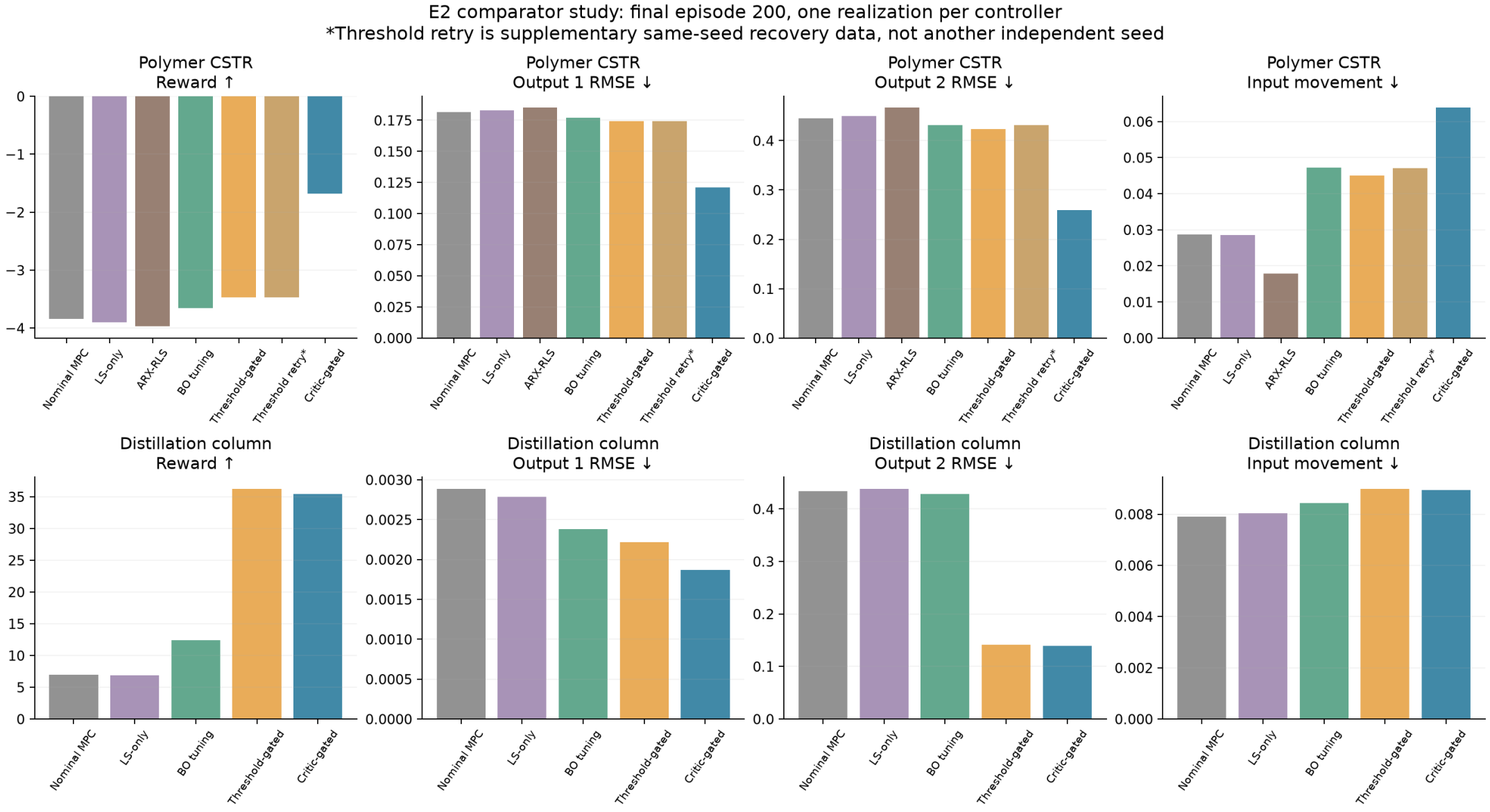

Combined controller at handoff: each connector is the same prescribed seed
Lower is better on every axis; nominal MPC is one common saved reference
Polymer CSTR: episodes 14–20
Viscosity RMSE (L/g)
Nominal MPC
Ungated RL
Critic-gated RL
Polymer CSTR: episodes 14–20
Temperature RMSE (K)
Ungated RL
Critic-gated RL
Polymer CSTR: episodes 14–20
Range-normalized input movement / step
Ungated RL
Critic-gated RL
Distillation column: episodes 14–20
Composition RMSE (mole fraction)
Nominal MPC
Ungated RL
Critic-gated RL
Distillation column: episodes 14–20
Temperature RMSE (native units)
Ungated RL
Critic-gated RL
Distillation column: episodes 14–20
Range-normalized input movement / step
Ungated RL
Critic-gated RL


Final deterministic evaluation (episode 300): report late performance separately
Dots are five seeds; MPC has one saved reference. Input activity is a cost, not a tracking metric.
Polymer CSTR
Reward ↑
Polymer CSTR
Viscosity RMSE ↓
Polymer CSTR
Temperature RMSE ↓
Polymer CSTR
Normalized movement ↓
Distillation column
Reward ↑
Distillation column
Composition RMSE ↓
Distillation column
Temperature RMSE ↓
Distillation column
Normalized movement ↓
MPC
RL
SG-RL

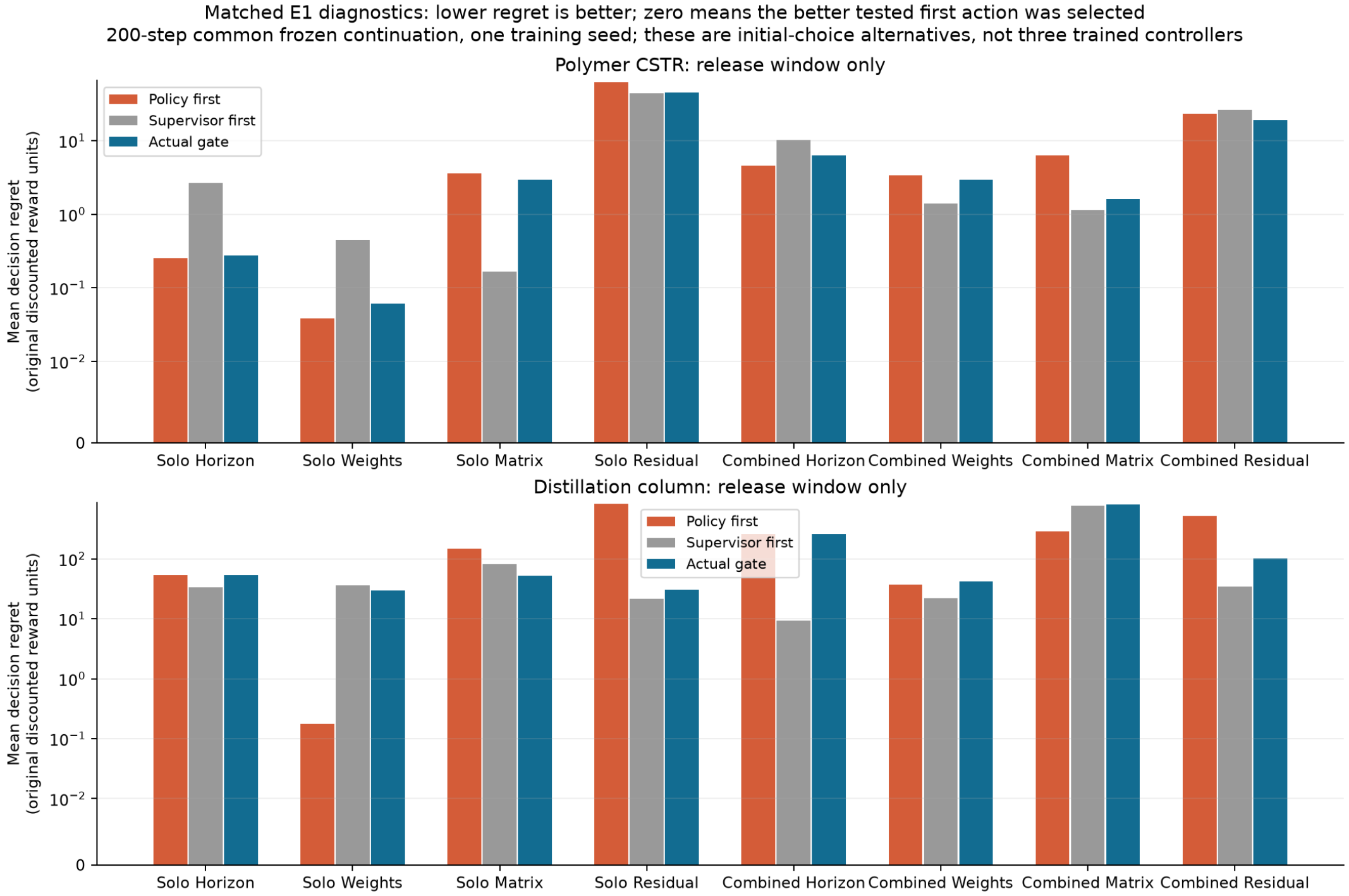
Column final outputs and inputs

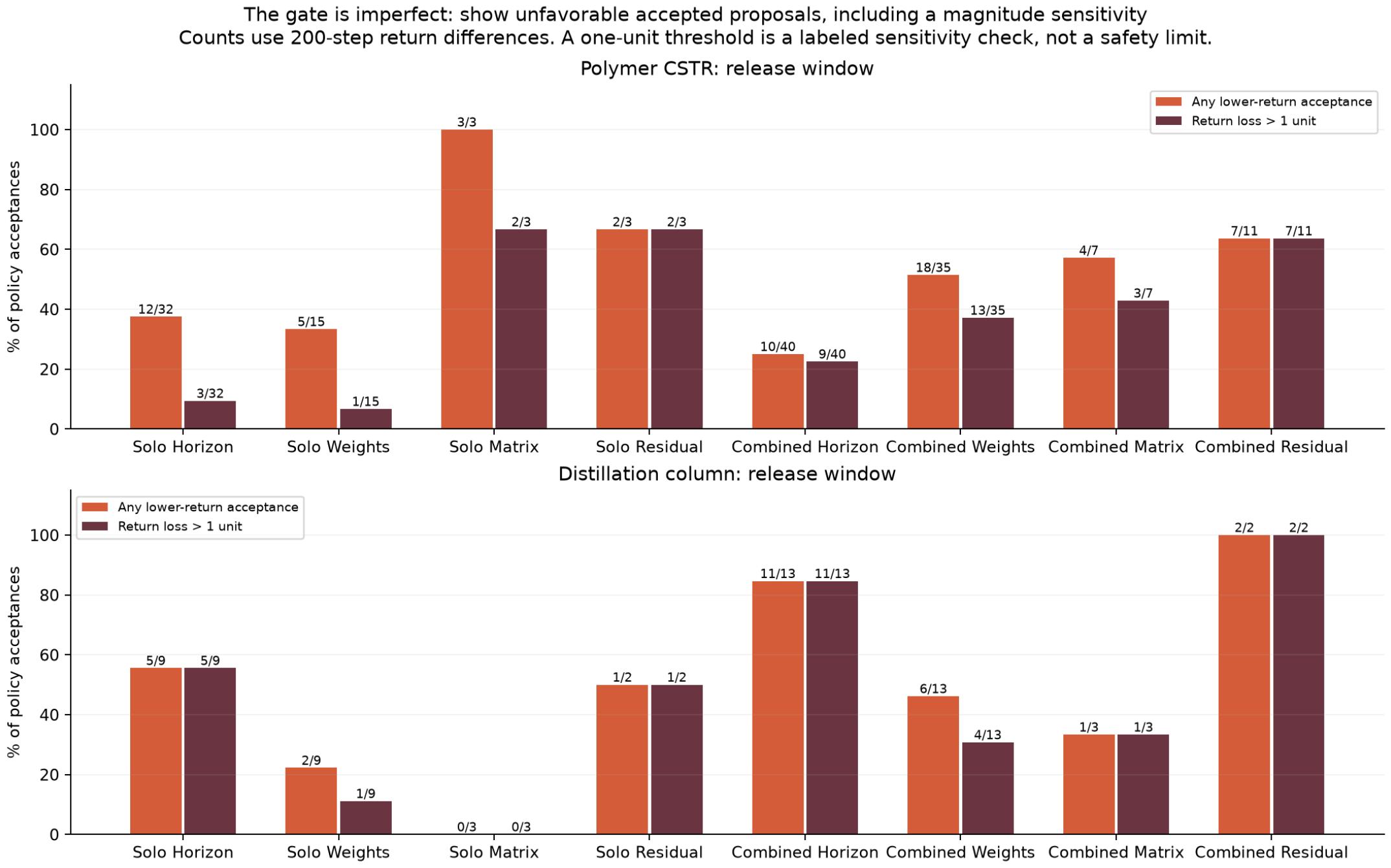



Polymer CSTR: combined-controller outputs and inputs, final episode
Mean ± sample SD across five seeds; baseline is one saved trajectory
Nominal MPC
Ungated RL
Critic-gated RL
Setpoint
Viscosity (L/g)
Reactor temperature (K)
Coolant flow (saved physical units)
Monomer flow (saved physical units)
Control step within episode 300


E2 comparator study: final episode 200, one realization per controller
*Threshold retry is supplementary same-seed recovery data, not another independent seed
Polymer CSTR
Reward ↑
Polymer CSTR
Output 1 RMSE ↓
Polymer CSTR
Output 2 RMSE ↓
Polymer CSTR
Input movement ↓
Distillation column
Reward ↑
Distillation column
Output 1 RMSE ↓
Distillation column
Output 2 RMSE ↓
Distillation column
Input movement ↓
Nominal MPC
LS-only
ARX-RLS
BO tuning
Threshold-gated
Threshold retry*
Critic-gated

E1 first-action regret at release

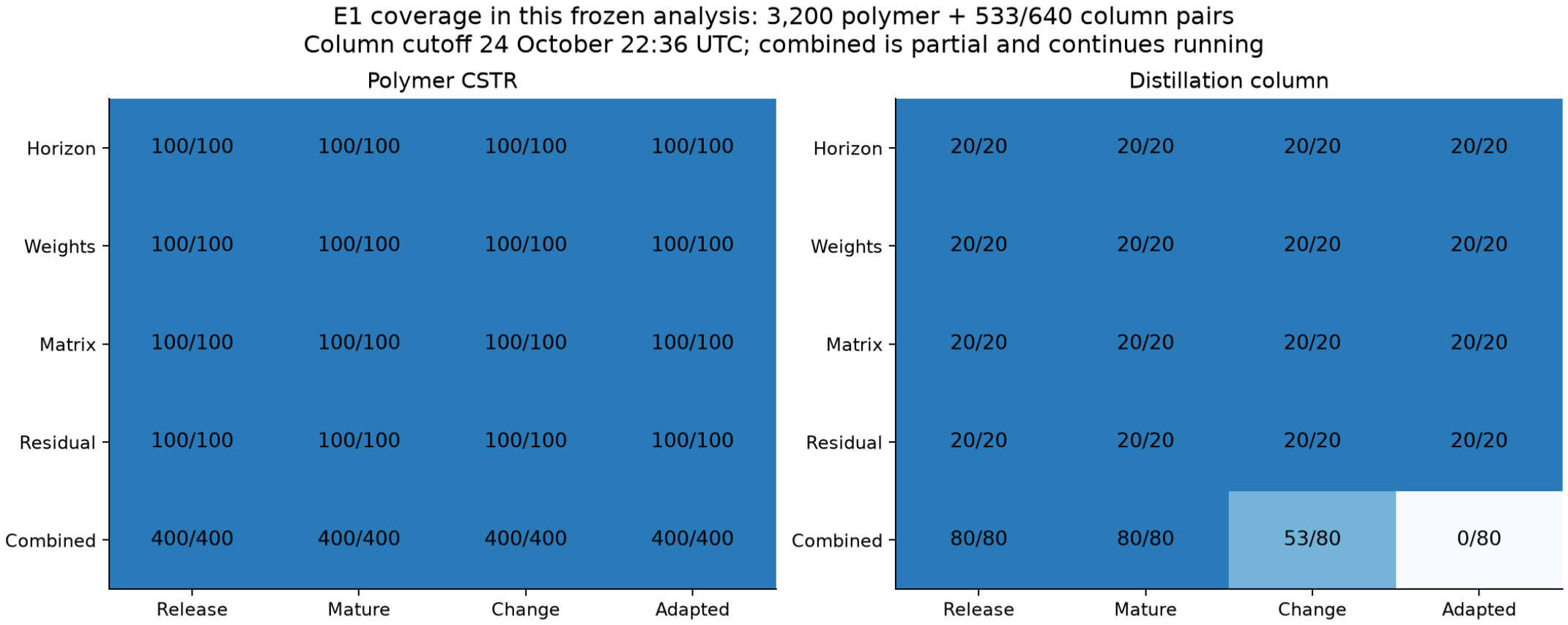


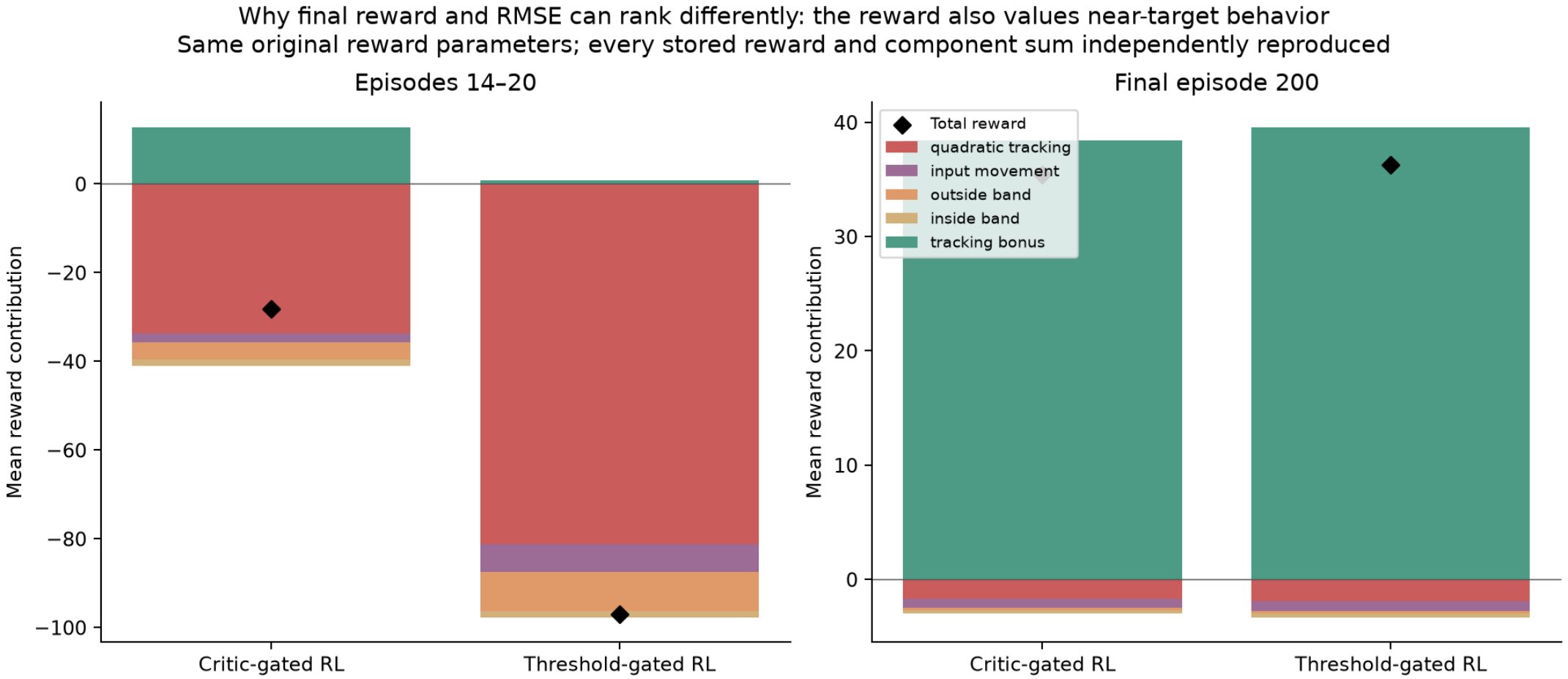

The gate is imperfect: show unfavorable accepted proposals, including a magnitude sensitivity
Counts use 200-step return differences. A one-unit threshold is a labeled sensitivity check, not a safety limit.
Polymer CSTR: release window
Any lower-return acceptance
Return loss > 1 unit
% of policy acceptances
0
20
40
60
80
100
12/32
3/32
5/15
1/15
3/3
2/3
2/3
2/3
10/40
9/40
18/35
13/35
4/7
3/7
7/11
7/11
Solo Horizon
Solo Weights
Solo Matrix
Solo Residual
Combined Horizon
Combined Weights
Combined Matrix
Combined Residual
Distillation column: release window
Any lower-return acceptance
Return loss > 1 unit
% of policy acceptances
0
20
40
60
80
100
5/9
5/9
2/9
1/9
0/3
0/3
1/2
1/2
11/13
11/13
6/13
4/13
1/3
1/3
2/2
2/2
Solo Horizon
Solo Weights
Solo Matrix
Solo Residual
Combined Horizon
Combined Weights
Combined Matrix
Combined Residual

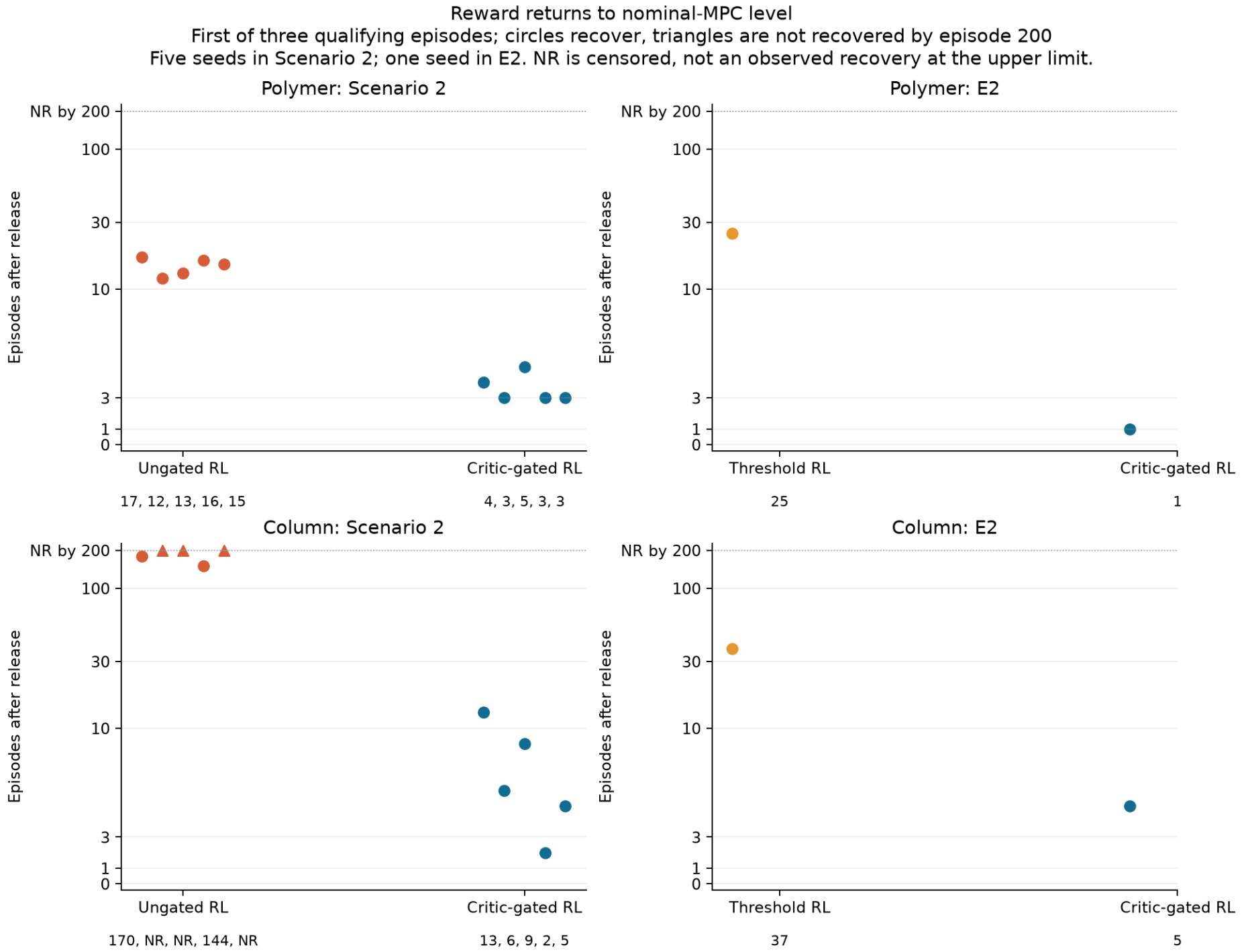
E3 held-out prediction ratios

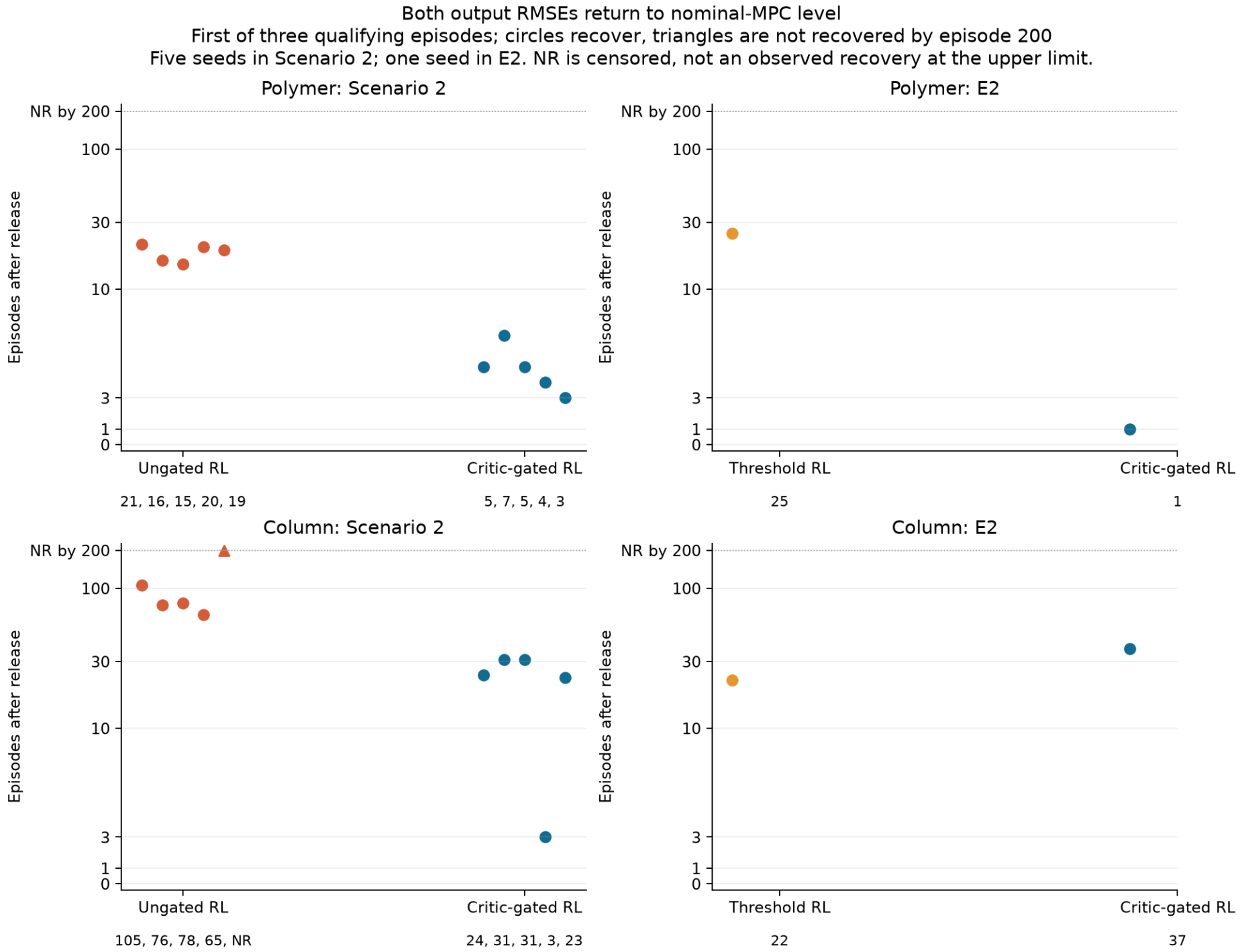


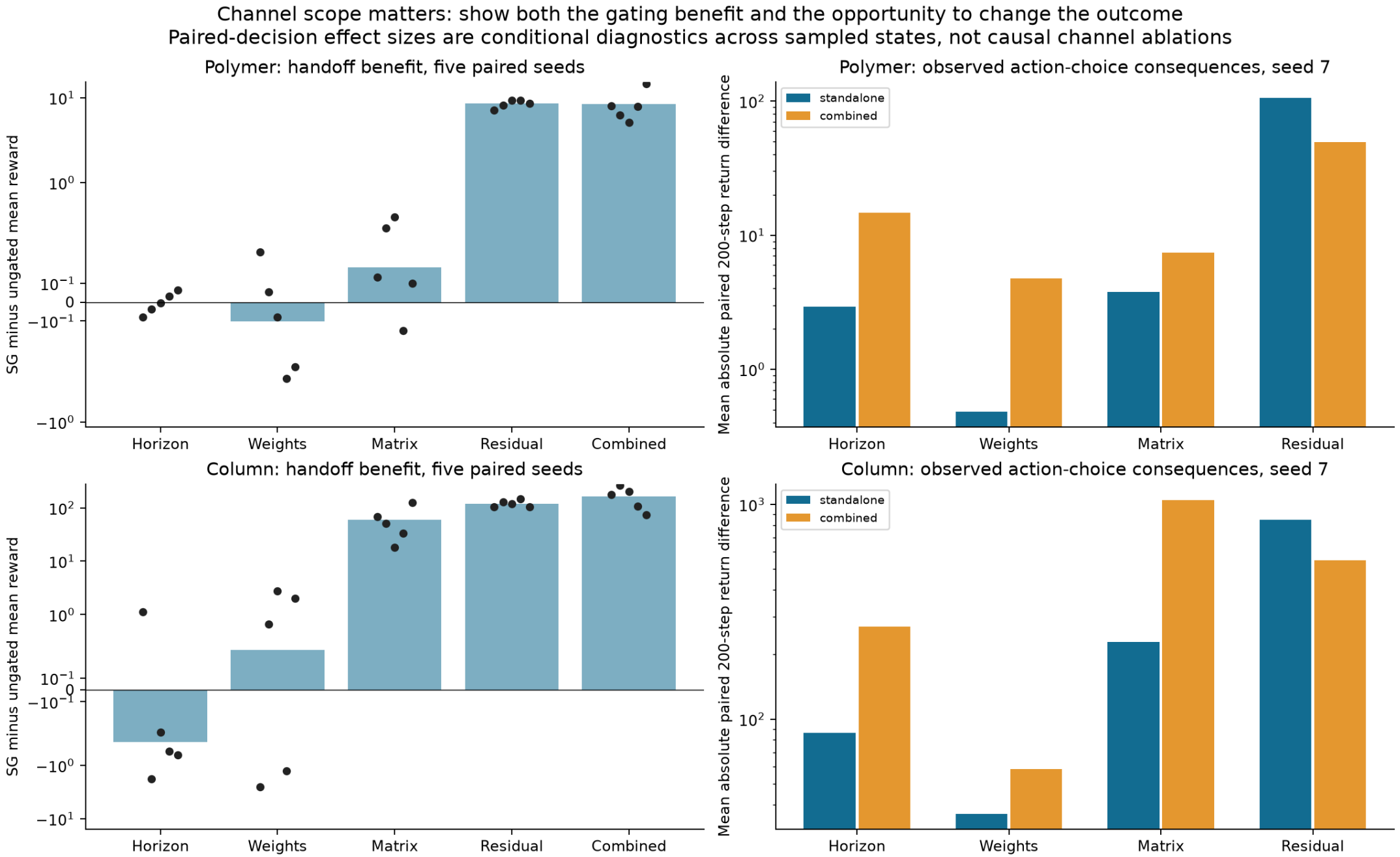
Complete polymer and partial column E1 coverage

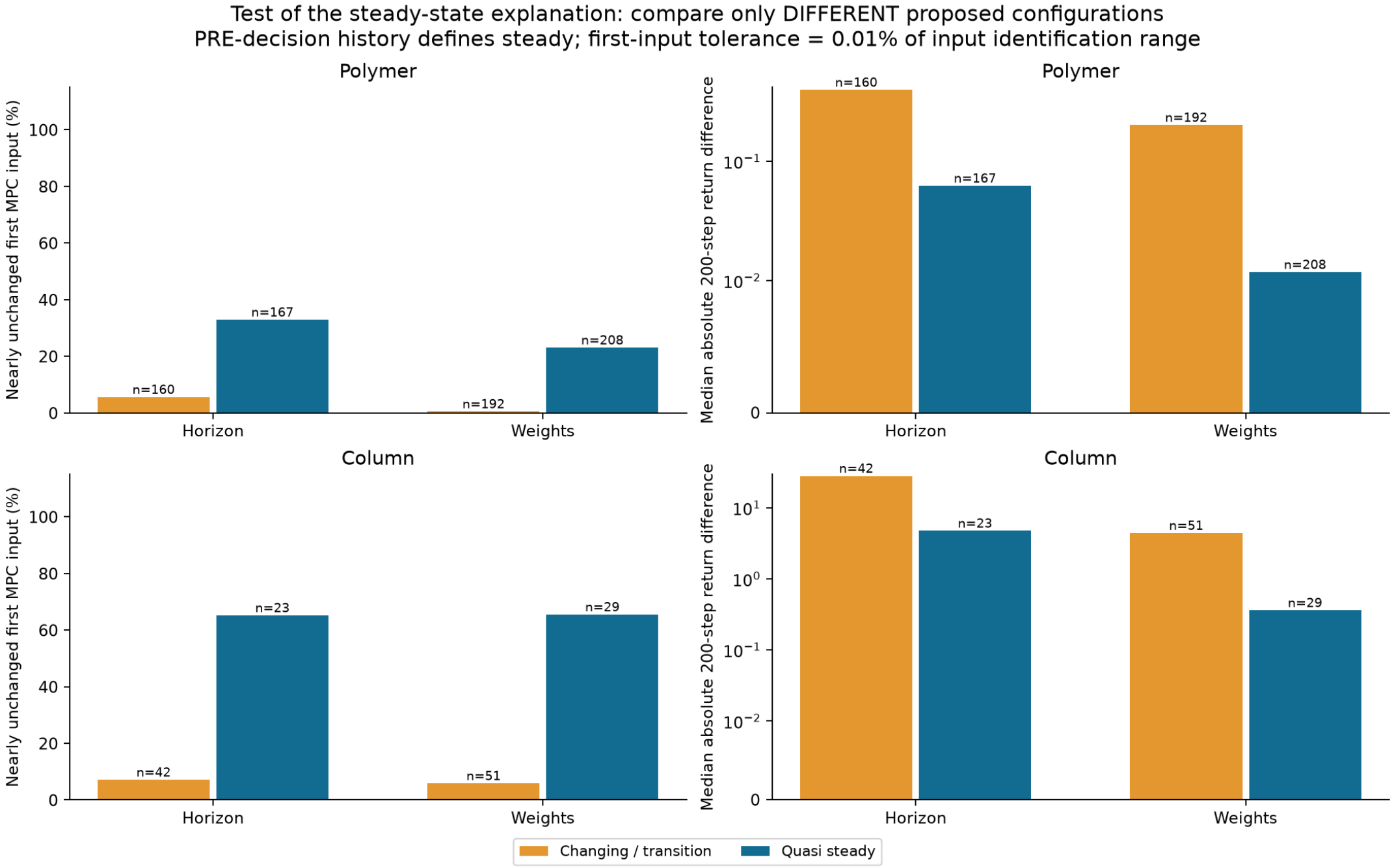


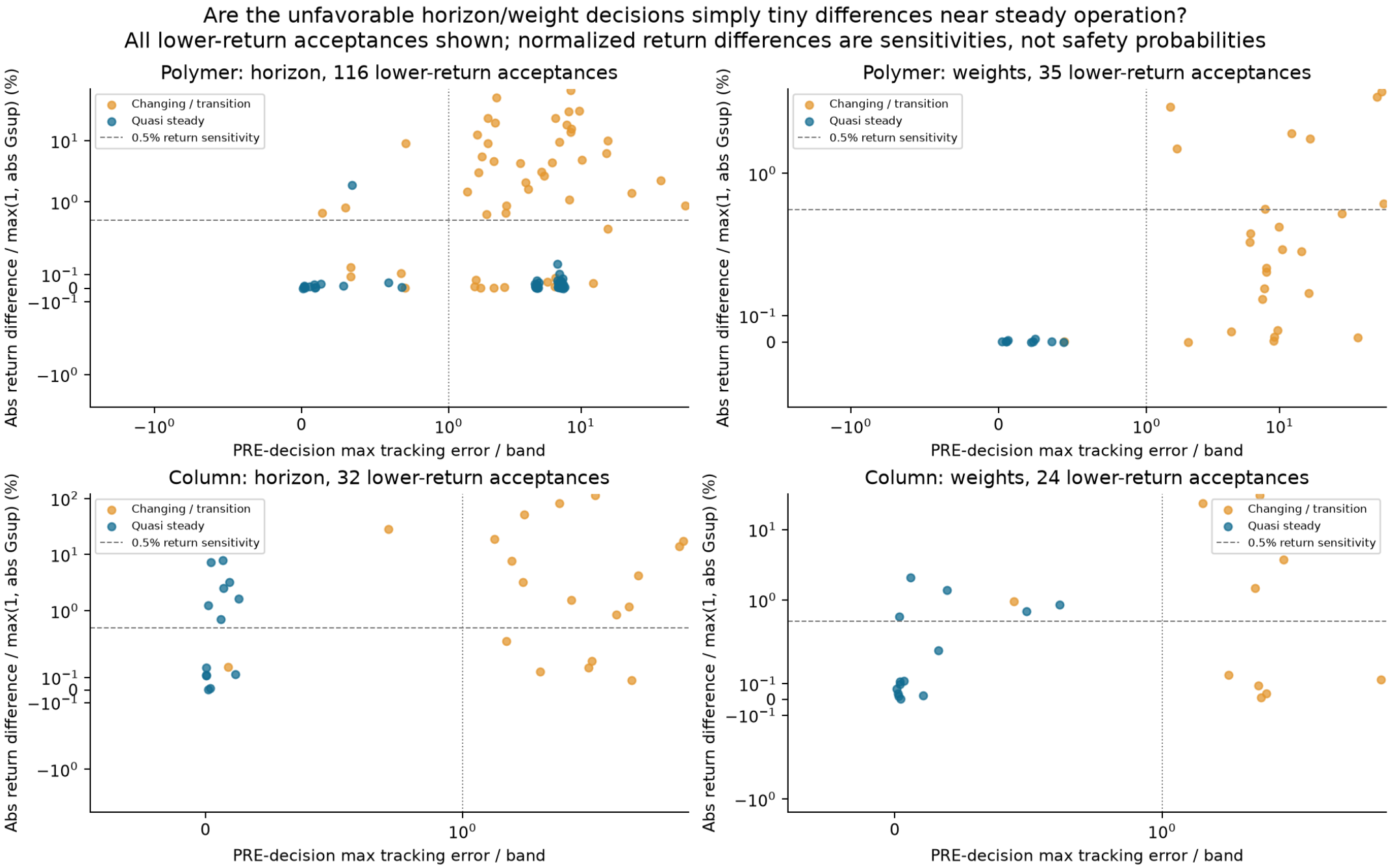

Why final reward and RMSE can rank differently: the reward also values near-target behavior
Same original reward parameters; every stored reward and component sum independently reproduced
Episodes 14–20
Final episode 200
Mean reward contribution
0
−20
−40
−60
−80
−100
40
30
20
10
Total reward
quadratic tracking
input movement
outside band
inside band
tracking bonus
Critic-gated RL
Threshold-gated RL

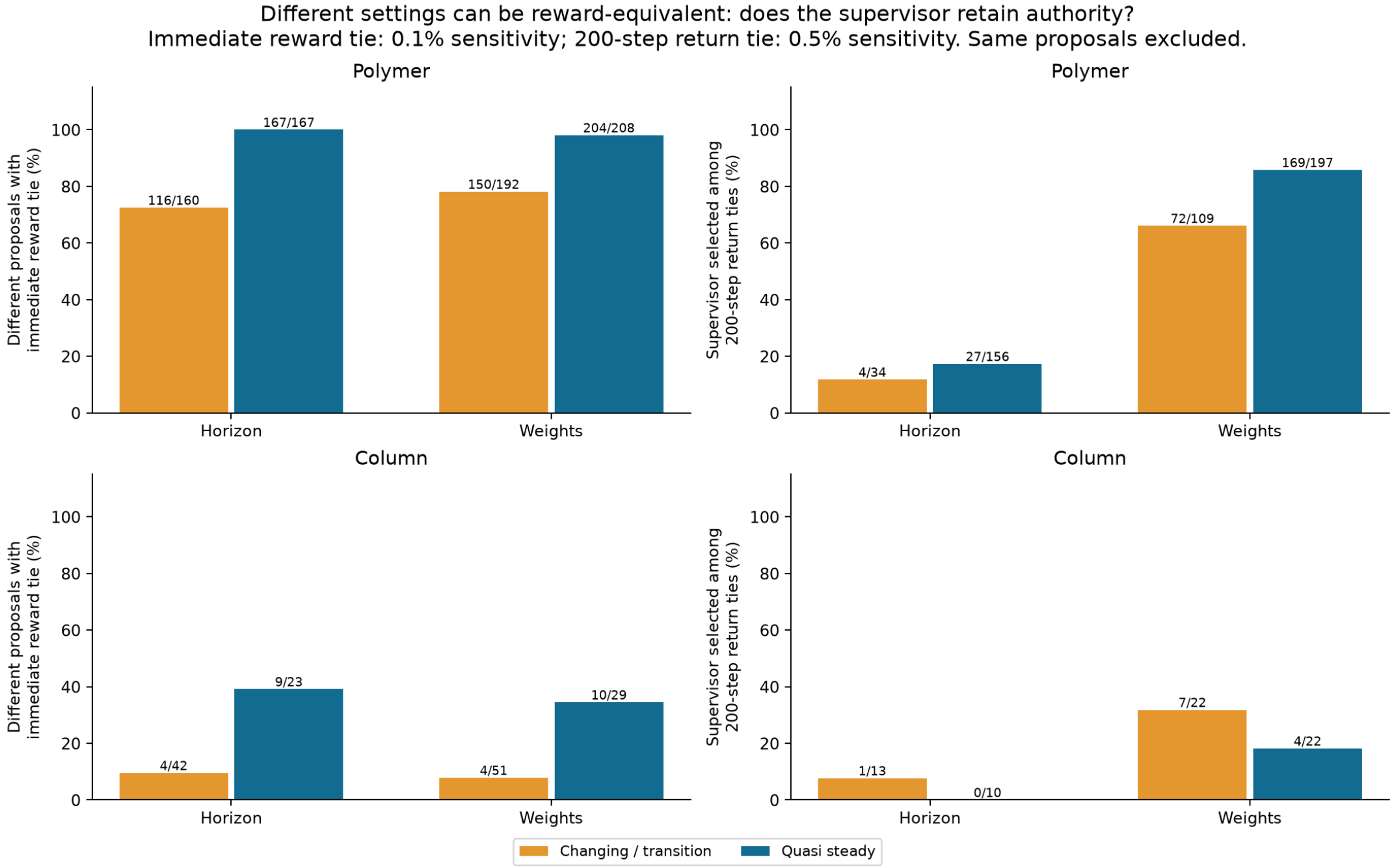

Reward returns to nominal-MPC level
First of three qualifying episodes; circles recover, triangles are not recovered by episode 200
Five seeds in Scenario 2; one seed in E2. NR is censored, not an observed recovery at the upper limit.
Polymer: Scenario 2
Polymer: E2
Column: Scenario 2
Column: E2
Episodes after release
NR by 200
100
30
10
3
1
0
Ungated RL
Critic-gated RL
Threshold RL
17, 12, 13, 16, 15
4, 3, 5, 3, 3
25
1
170, NR, NR, 144, NR
13, 6, 9, 2, 5
37
5


Both output RMSEs return to nominal-MPC level
First of three qualifying episodes; circles recover, triangles are not recovered by episode 200
Five seeds in Scenario 2; one seed in E2. NR is censored, not an observed recovery at the upper limit.
Polymer: Scenario 2
Polymer: E2
Column: Scenario 2
Column: E2
Episodes after release
NR by 200
100
30
10
3
1
0
Ungated RL
Critic-gated RL
Threshold RL
21, 16, 15, 20, 19
5, 7, 5, 4, 3
25
1
105, 76, 78, 65, NR
24, 31, 31, 3, 23
22
37


Channel scope matters: show both the gating benefit and the opportunity to change the outcome
Paired-decision effect sizes are conditional diagnostics across sampled states, not causal channel ablations
Polymer: handoff benefit, five paired seeds
SG minus ungated mean reward
Horizon
Weights
Matrix
Residual
Combined
Polymer: observed action-choice consequences, seed 7
Mean absolute paired 200-step return difference
standalone
combined
Column: handoff benefit, five paired seeds
Column: observed action-choice consequences, seed 7


Test of the steady-state explanation: compare only DIFFERENT proposed configurations
PRE-decision history defines steady; first-input tolerance = 0.01% of input identification range
Polymer
Nearly unchanged first MPC input (%)
n=160
n=167
n=192
n=208
Horizon
Weights
Polymer
Median absolute 200-step return difference
n=160
n=167
n=192
n=208
Horizon
Weights
Column
Nearly unchanged first MPC input (%)
n=42
n=23
n=51
n=29
Horizon
Weights
Column
Median absolute 200-step return difference
n=42
n=23
n=51
n=29
Horizon
Weights
Changing / transition
Quasi steady


Are the unfavorable horizon/weight decisions simply tiny differences near steady operation?
All lower-return acceptances shown; normalized return differences are sensitivities, not safety probabilities
Polymer: horizon, 116 lower-return acceptances
Polymer: weights, 35 lower-return acceptances
Column: horizon, 32 lower-return acceptances
Column: weights, 24 lower-return acceptances
Changing / transition
Quasi steady
0.5% return sensitivity
Abs return difference / max(1, abs Gsup) (%)
PRE-decision max tracking error / band


Different settings can be reward-equivalent: does the supervisor retain authority?
Immediate reward tie: 0.1% sensitivity; 200-step return tie: 0.5% sensitivity. Same proposals excluded.
Polymer
Different proposals with immediate reward tie (%)
116/160
167/167
150/192
204/208
Horizon
Weights
Polymer
Supervisor selected among 200-step return ties (%)
4/34
27/156
72/109
169/197
Horizon
Weights
Column
Different proposals with immediate reward tie (%)
4/42
9/23
4/51
10/29
Horizon
Weights
Column
Supervisor selected among 200-step return ties (%)
1/13
0/10
7/22
4/22
Horizon
Weights
Changing / transition
Quasi steady